LEVEL 1 PRACTICE PROBLEMS:

QUESTION:

1. Write a program to check if a number is divisible by 5

I/P => number

O/P => Is the number divisible by 5?

SOLUTION:

PROGRAM NAME: Divisible by 5 program

```
import java.util.Scanner;

public class DivisibleByFive {

    public static void main(String[] args) {

        Scanner input = new Scanner(System.in);

        int number = 0;

        System.out.print("Enter a number: ");
        number = input.nextInt();

        boolean isDivisibleByFive = (number % 5 == 0);

        if (isDivisibleByFive) {
```

```
            System.out.println("The number is divisible by 5.");
        }
         else {
            System.out.println("The number is NOT divisible by
5.");
        }

        input.close();
    }
}
```

SAMPLE OUTPUT:

Output

```
Enter a number: 25
The number is divisible by 5.

=== Code Execution Successful ===
```

QUESTION:

2. Write a program to check if the first is the smallest of the 3 numbers.

I/P => number1, number2, number3

O/P => Is the first number the smallest?

SOLUTION:

PROGRAM NAME: First Number is smallest of three

```
import java.util.Scanner;
public class FirstIsSmallest {

    public static void main(String[] args) {

        Scanner input = new Scanner(System.in);

        int number1 = 0;
        int number2 = 0;
        int number3 = 0;

        System.out.print("Enter first number: ");
        number1 = input.nextInt();

        System.out.print("Enter second number: ");
        number2 = input.nextInt();
```

```
        System.out.print("Enter third number: ");
        number3 = input.nextInt();

        boolean isFirstSmallest = (number1 < number2) &&
(number1 < number3);

        if (isFirstSmallest) {
            System.out.println("The first number is the smallest.");
        } else {
            System.out.println("The first number is NOT the
smallest.");
        }
        input.close();
    }
}
```

SAMPLE OUTPUT:

Output

```
Enter first number: 10
Enter second number: 25
Enter third number: 30
The first number is the smallest.

=== Code Execution Successful ===
```

QUESTION:

3. Write a program to check if the first, second, or third number is the largest of the three.

I/P => number1, number2, number3

O/P =>

Is the first number the largest?

Is the second number the largest?

Is the third number the largest?

SOLUTION:

PROGRAM NAME: Check which number is largest

```
import java.util.Scanner;
public class LargestOfThree {

    public static void main(String[] args) {

        Scanner input = new Scanner(System.in);

        int number1 = 0;
        int number2 = 0;
        int number3 = 0;

        System.out.print("Enter first number: ");
        number1 = input.nextInt();
```

```
        System.out.print("Enter second number: ");
        number2 = input.nextInt();

        System.out.print("Enter third number: ");
        number3 = input.nextInt();

        boolean isFirstLargest = (number1 > number2) && (number1 > number3);
        boolean isSecondLargest = (number2 > number1) && (number2 > number3);
        boolean isThirdLargest = (number3 > number1) && (number3 > number2);

        if (isFirstLargest) {
            System.out.println("The first number is the largest.");
        }
        else if (isSecondLargest) {
            System.out.println("The second number is the largest.");
        }
        else if (isThirdLargest) {
            System.out.println("The third number is the largest.");
        }
        else {
            System.out.println("Two or more numbers are equal.");
        }
        input.close();
    }
```

}

SAMPLE OUTPUT:

Output

```
Enter first number: 45
Enter second number: 12
Enter third number: 30
The first number is the largest.

=== Code Execution Successful ===
```

LEVEL 2 PRACTICE PROBLEMS:

QUESTION:

1. Create a program to print odd and even numbers between 1 to the number entered by the user.

Hint =>

a. Get an integer input from the user, assign to a variable number and check for Natural Number

b. Using a for loop, iterate from 1 to the number

c. In each iteration of the loop, print the number is odd or even number

SOLUTION:

PROGRAM NAME: Print Odd and Even numbers from 1 to N

```
import java.util.Scanner;

public class OddEvenPrinter {

    public static void main(String[] args) {

        Scanner input = new Scanner(System.in);

        int number = 0;

        System.out.print("Enter a natural number: ");
        number = input.nextInt();

        if (number <= 0) {
            System.out.println("Please enter a valid natural number greater than 0.");
        } else {
```

```
        for (int i = 1; i <= number; i++) {

            // Check if the current number is even or odd
            if (i % 2 == 0) {
                System.out.println(i + " is an Even number");
            } else {
                System.out.println(i + " is an Odd number");
            }
        }
    }

    input.close();
  }
}
```

SAMPLE OUTPUT:

Output

```
Enter a natural number: 5
1 is an Odd number
2 is an Even number
3 is an Odd number
4 is an Even number
5 is an Odd number

=== Code Execution Successful ===
```

QUESTION:

2. Create a program to find the bonuses of employees based on their years of service.

Hint =>

a. Zara decided to give a bonus of 5% to employees whose year of service is more than 5 years.

b. Take salary and year of service in the year as input.

c. Print the bonus amount.

SOLUTION:

PROGRAM NAME: Calculate bonus based on years of service

```
import java.util.Scanner;

public class EmployeeBonusCalculator {

    public static void main(String[] args) {

        Scanner input = new Scanner(System.in);

        double salary = 0.0;
```

```
        int yearsOfService = 0;
        double bonusPercentage = 0.0;
        double bonusAmount = 0.0;

        System.out.print("Enter your salary: ");
        salary = input.nextDouble();

        System.out.print("Enter your years of service: ");
        yearsOfService = input.nextInt();

        if (yearsOfService > 5) {
            bonusPercentage = 0.05; // 5% bonus
            bonusAmount = salary * bonusPercentage;
            System.out.println("You are eligible for a bonus.");
        } else {
            bonusAmount = 0.0;
            System.out.println("You are not eligible for a bonus.");
        }

        System.out.println("Bonus Amount: " + bonusAmount);

        input.close();
    }
}
```

SAMPLE OUTPUT:

Output

```
Enter your salary: 5000
Enter your years of service: 6
You are eligible for a bonus.
Bonus Amount: 250.0

=== Code Execution Successful ===
```

LEVEL 3 PRACTICE PROBLEMS:

QUESTION:

1. Write a LeapYear program that takes a year as input and outputs the Year is a Leap Year or not a Leap Year.

   Hint =>

   a. The LeapYear program only works for year >= 1582, corresponding to a year in the Gregorian calendar. So ensure to check for the same.

   b. Further, the Leap Year is a Year divisible by 4 and not 100 unless it is divisible by 400. E.g. 1800 is not a Leap Year and 2000 is a Leap Year.

c. Write code having multiple if else statements based on conditions provided above and a second part having only one if statement and multiple logical

SOLUTION:

PROGRAM NAME: Leap year using multiple if statements

```
import java.util.Scanner;

public class LeapYearIfElse {

    public static void main(String[] args) {

        Scanner input = new Scanner(System.in);

        int year = 0;

        System.out.print("Enter a year: ");
        year = input.nextInt();

        if (year < 1582) {
            System.out.println("Leap Year calculation is valid only for year 1582 and above.");
        }
        else {
```

```
            if (year % 400 == 0) {
                System.out.println(year + " is a Leap Year.");
            }
            else if (year % 100 == 0) {
                System.out.println(year + " is NOT a Leap Year.");
            }
            else if (year % 4 == 0) {
                System.out.println(year + " is a Leap Year.");
            }
            else {
                System.out.println(year + " is NOT a Leap Year.");
            }
        }

        input.close();
    }
}
```

SAMPLE OUTPUT:

Output

```
Enter a year: 2000
2000 is a Leap Year.

=== Code Execution Successful ===
```

Output

```
Enter a year: 1800
1800 is NOT a Leap Year.

=== Code Execution Successful ===
```

Output

```
Enter a year: 1400
Leap Year calculation is valid only for year 1582 and above.

=== Code Execution Successful ===
```

Program 2: Leap Year (Using One if with Logical Operators)

```
import java.util.Scanner;

public class LeapYearLogical {

    public static void main(String[] args) {

        Scanner input = new Scanner(System.in);

        int year = 0;

        System.out.print("Enter a year: ");
        year = input.nextInt();

        if (year < 1582) {
            System.out.println("Leap Year calculation is valid only for year 1582
and above.");
        }
        else {

            // Leap year logic using a single if statement with logical operators
            if ((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0)) {
                System.out.println(year + " is a Leap Year.");
            }
            else {
                System.out.println(year + " is NOT a Leap Year.");
            }
        }
```

```
        input.close();
    }
}
```

SAMPLE OUTPUT:

Output

```
Enter a year: 2024
2024 is a Leap Year.

=== Code Execution Successful ===
```

Output

```
Enter a year: 1900
1900 is NOT a Leap Year.

=== Code Execution Successful ===
```